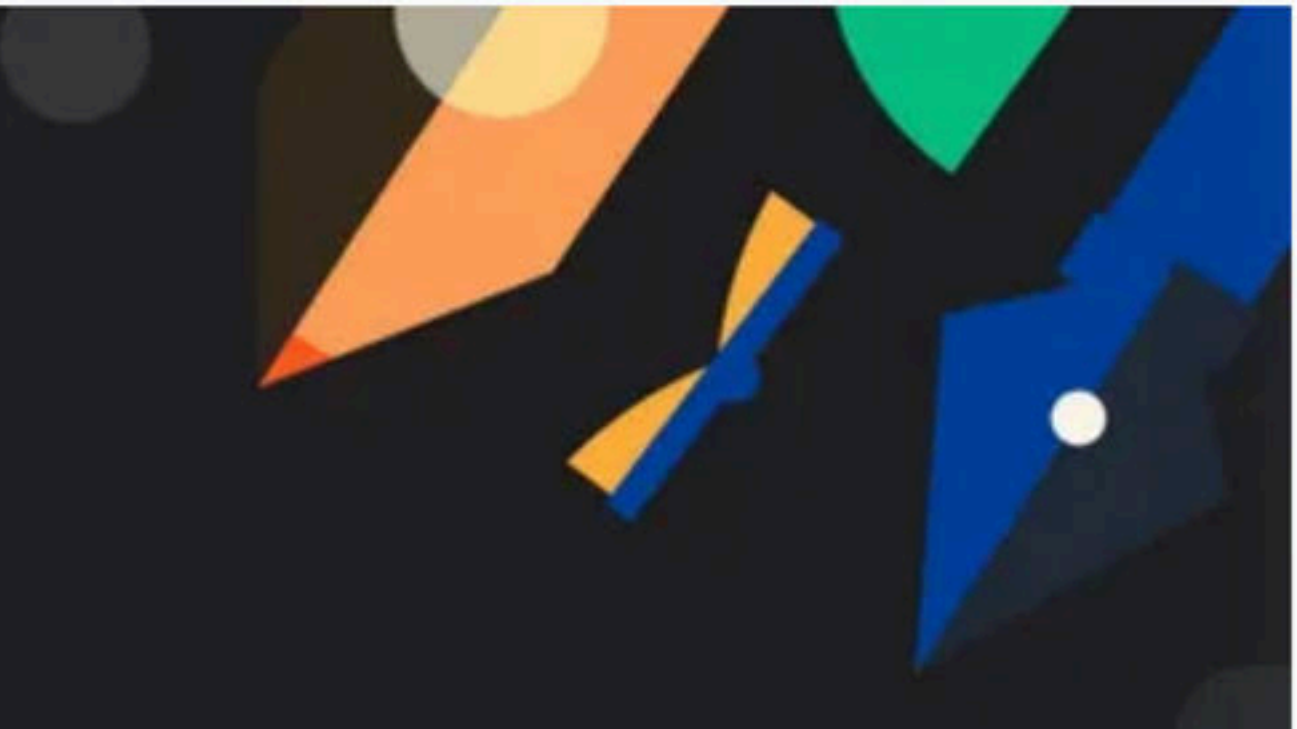




Mega Class: BST

Special class

①

LC 783

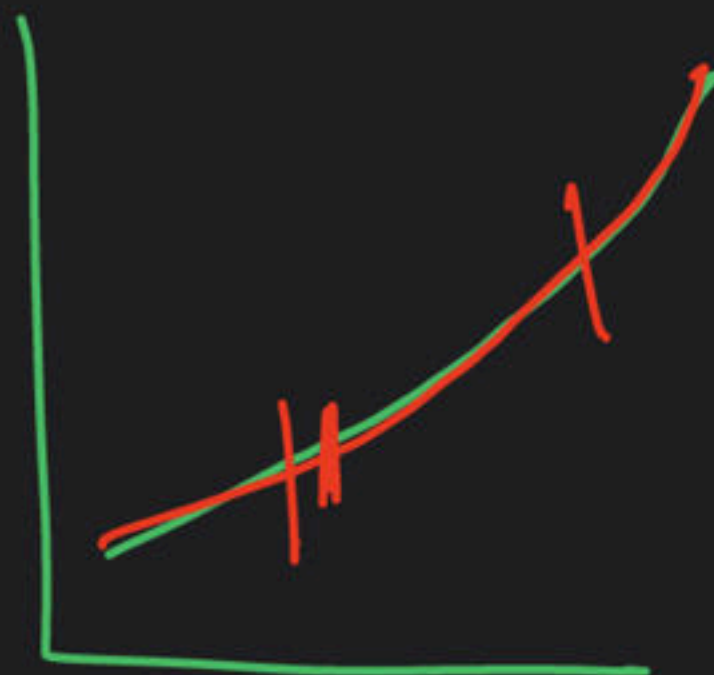
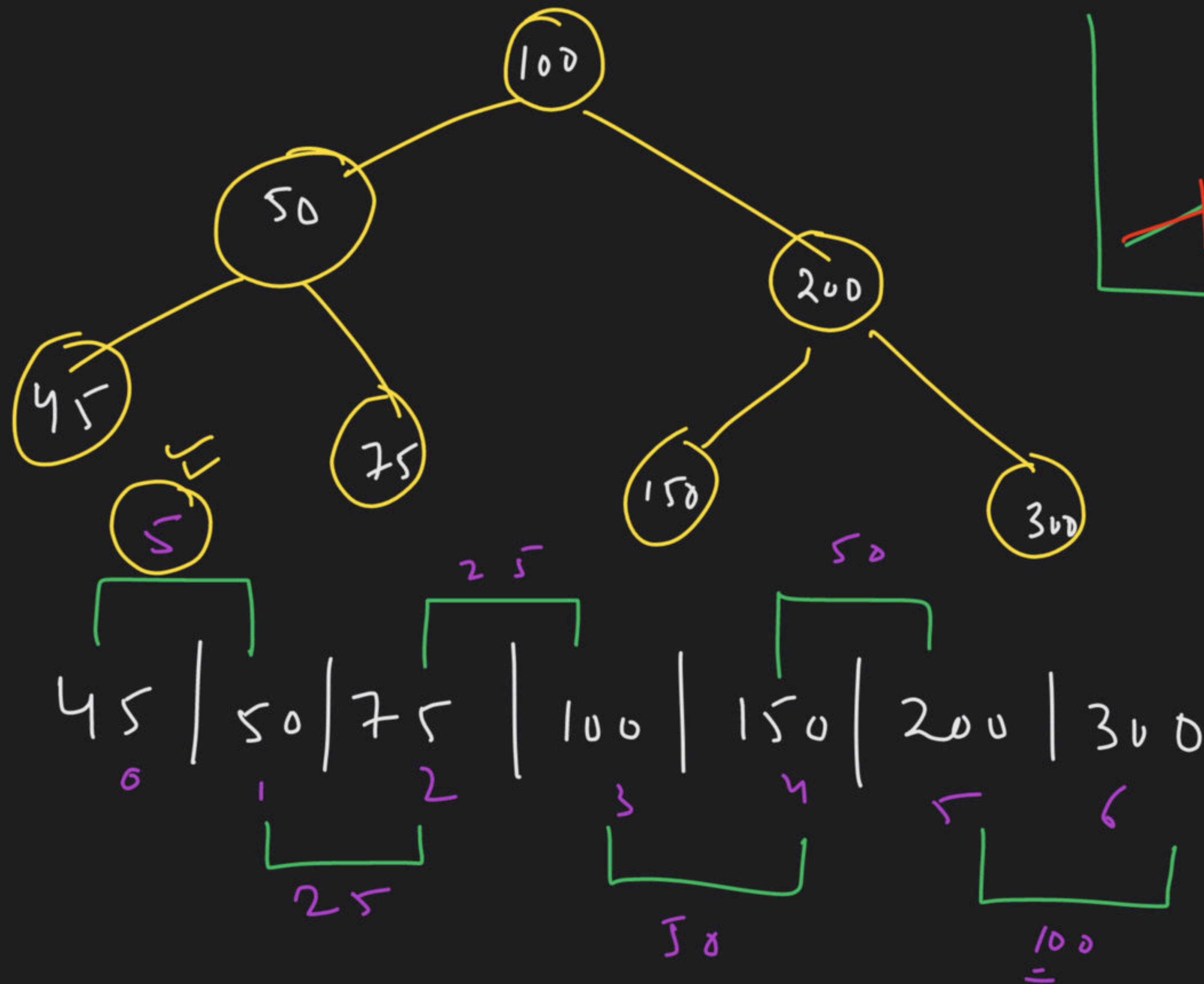
Min Distance b/w BST
Nodes
=

MI

Inorder T $\rightarrow$ vector

find min diff
b/w any two
nos.

MI



→ in order

Vector =>

find min of adjacent diff



①

inorder $\rightarrow O(n)$

②

store $\rightarrow$ SC: $O(n)$

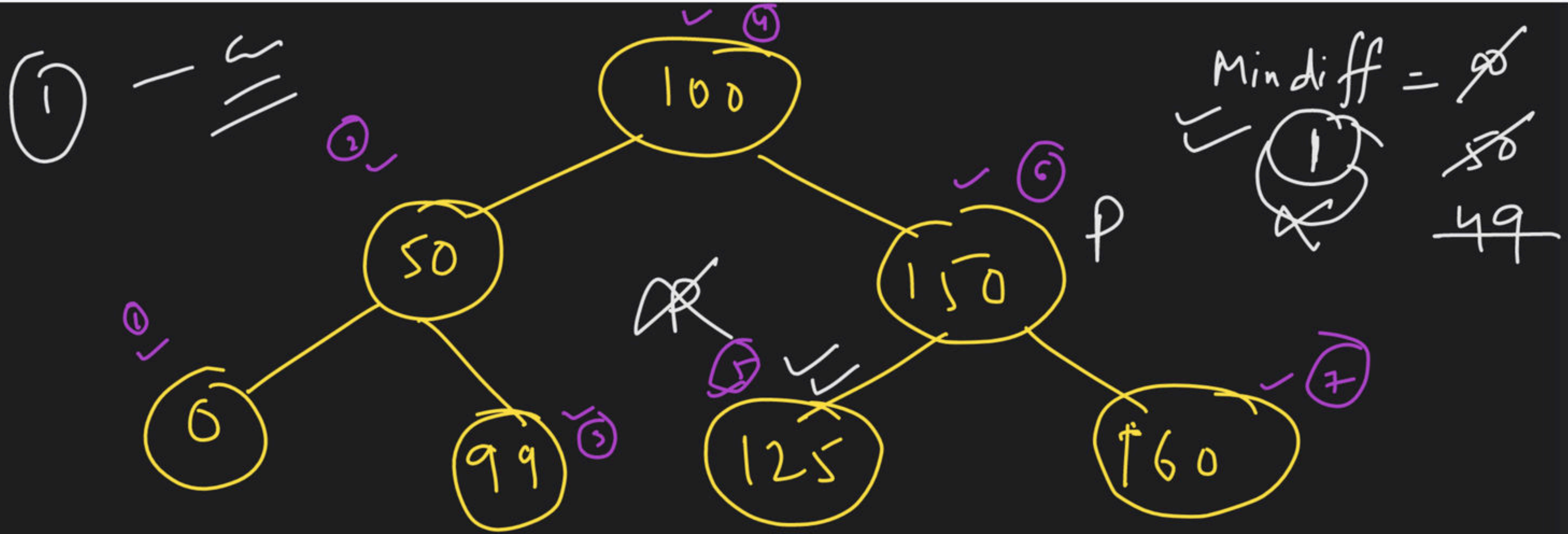
③

Vector loop $\rightarrow$ $O(n)$

(M2)

Avoid using extra ^{vector} space

→ Live ^{in order} Traversal → 4 find the min diff.



Prev $\Rightarrow$ Null
=

① 150 $\Rightarrow$ 150 - 125 = 25
⑦ 160 $\Rightarrow$ 160 - 150 = 10

① 0 $\Rightarrow$ ~~X~~ $\rightarrow$ prev = 0
② 50 $\Rightarrow$ 50 - 0 $\Rightarrow$ 50 $\Rightarrow$ Prev
③ 99 $\Rightarrow$ 99 - 50 $\Rightarrow$ 49 $\Rightarrow$ Prev $\Rightarrow$ 99
④ 100 $\Rightarrow$ 100 - 99 $\Rightarrow$ 1
⑤ 125 $\Rightarrow$ 125 - 100 $\Rightarrow$ 25
prev = 100
max = 125

② LC 98: Validate BST

5 min Think 4 Read

8:37 pm

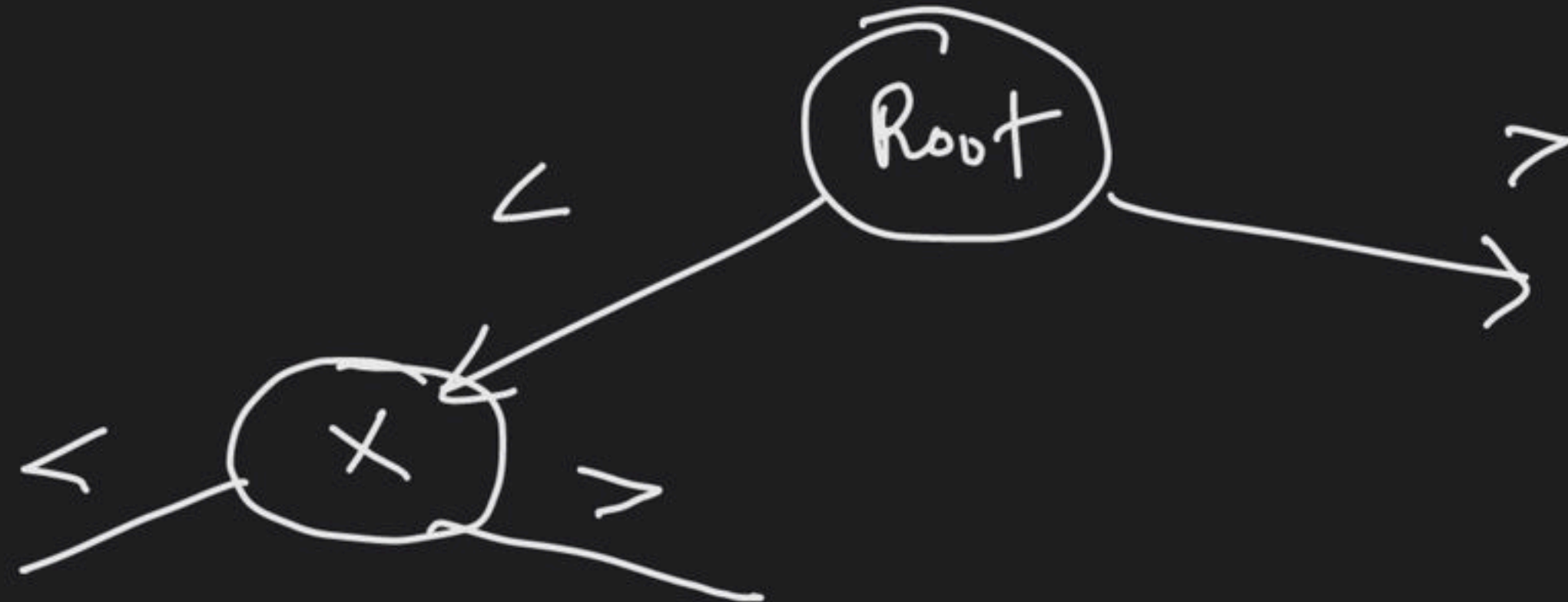
M1

BST $\rightarrow$ inorder $\rightarrow$ sorted
Brute force

M2

Range Checking

BST property



①

element check

$[-\infty, \infty)$

T_{true}

lb
 $\Downarrow$
 $-\infty$

ub
 $\Downarrow$
 ∞

$(-\infty, 100)$
 $\Downarrow$

$(100, \infty)$

$(200, \infty)$

$(-\infty, 0)$
 $\Downarrow$

$(0, 100)$
 $\Downarrow$

$(100, 200)$

$(200, 300)$

$(300, \infty)$

$(-\infty, -50)$
 $\Downarrow$

$(-50, 0)$
 $\Downarrow$

$(0, 75)$

$(100, 200)$

$(200, 300)$

$(300, \infty)$

$(-\infty, -50)$
 $\Downarrow$

$(-50, 0)$
 $\Downarrow$

$(0, 75)$

$(100, 200)$

$(200, 300)$

$(300, \infty)$

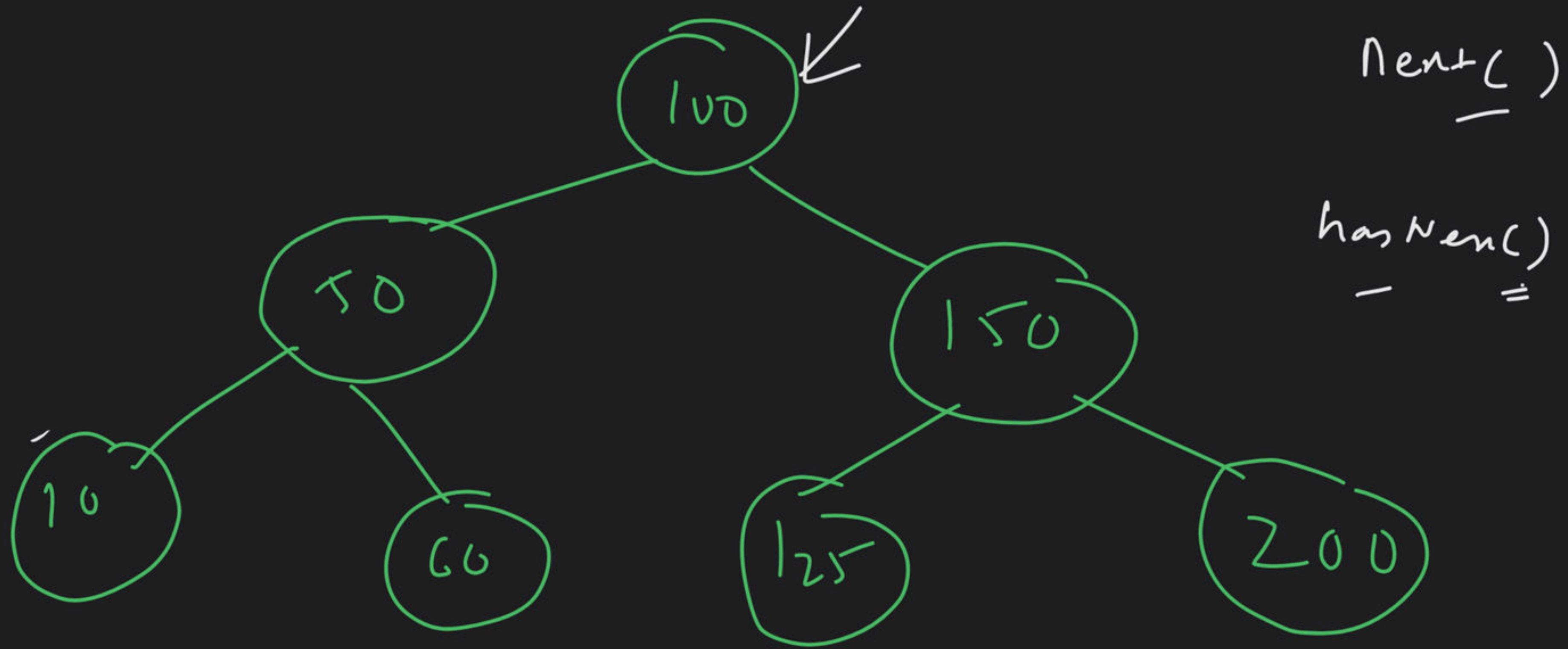
③ LC 173: BST iterator

✓✓

⇒ 9:07

Think ahead

→



① Next() $\Rightarrow$ 10

② Next() $\Rightarrow$ 50

③ Next() $\Rightarrow$ 60

(M1)

Vector =)

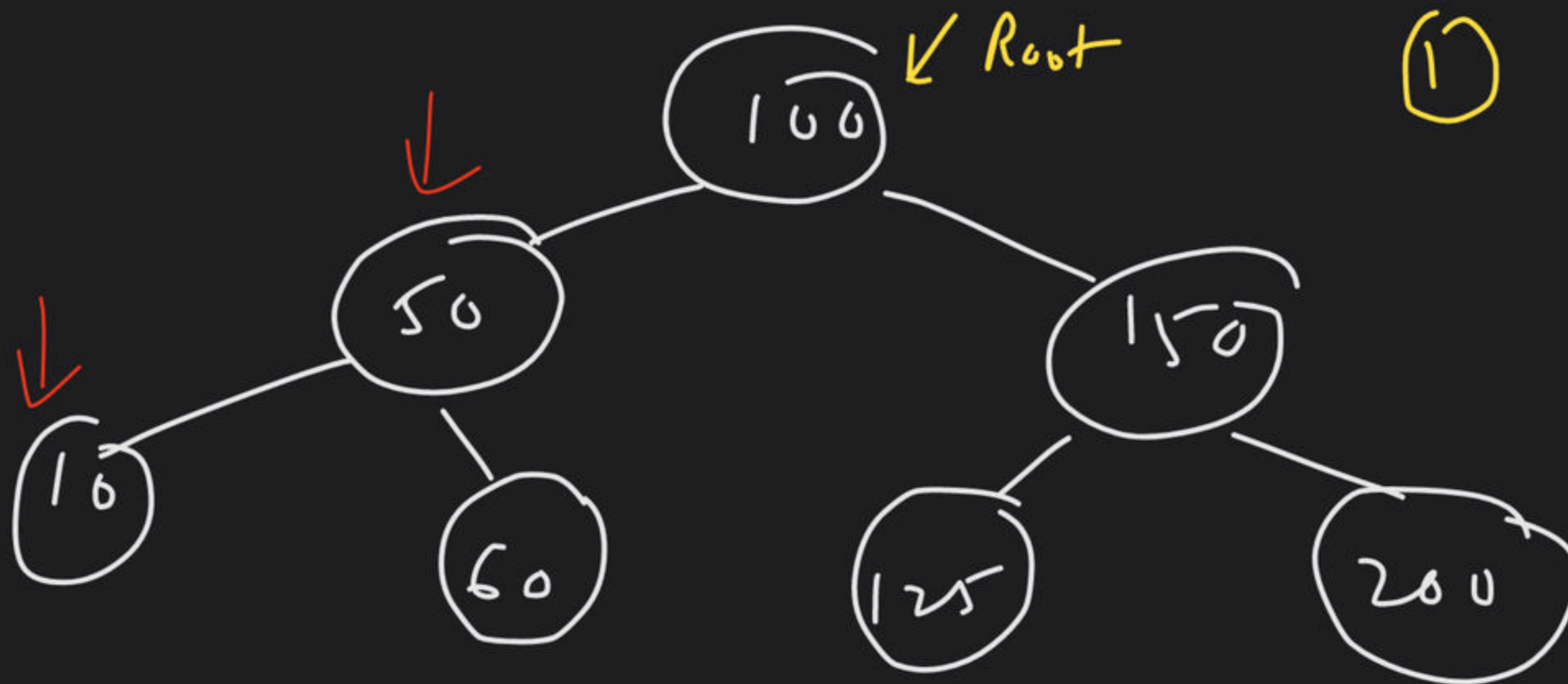
10	50	60	100	125	150	200
0	1	2	3	4	5	6

i

① next () => 10 , i++

② hasNext () => $i \geq 7$ → false
else True

M2



① push all
leftmost nodes
=

Stack

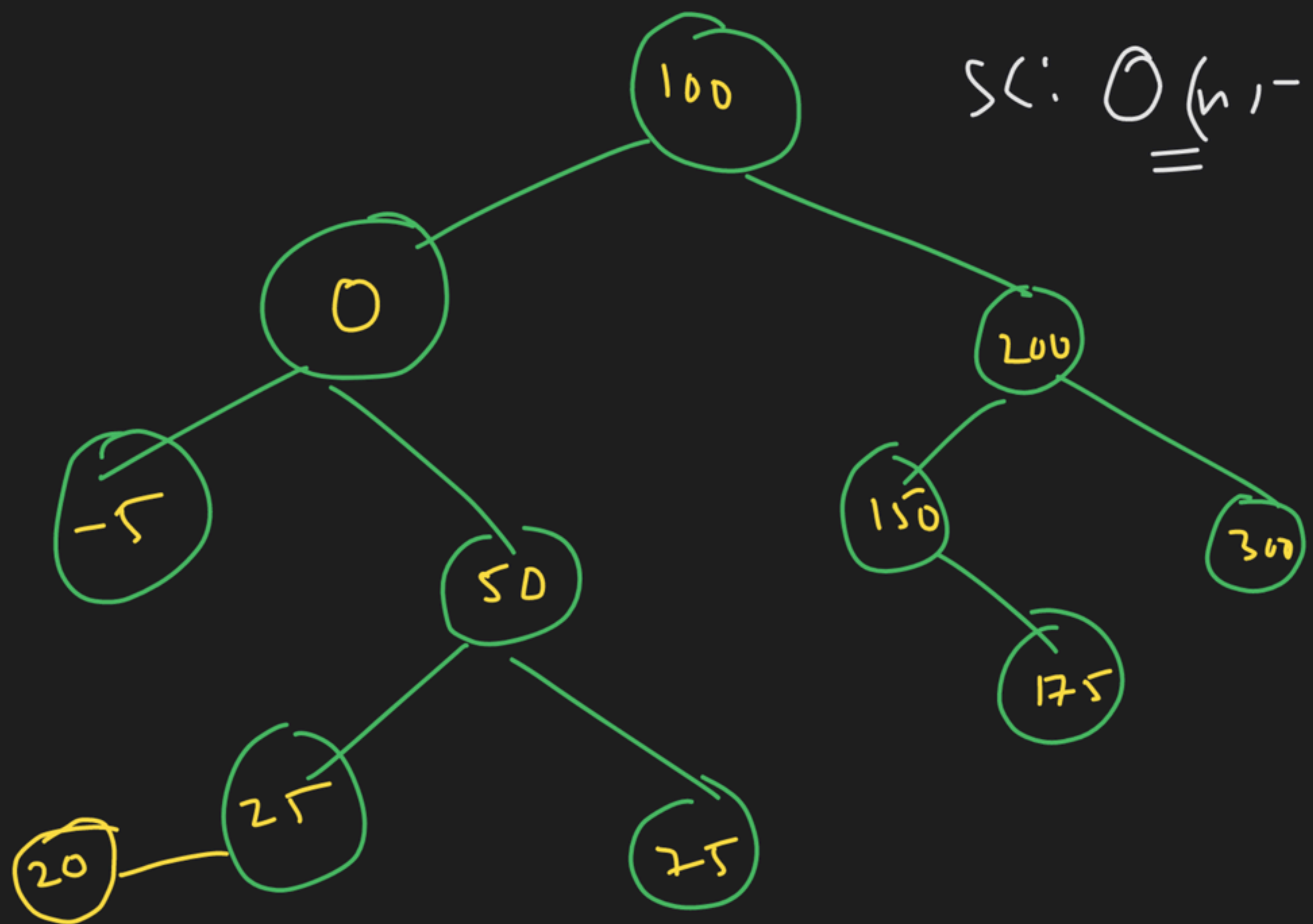
next() $\Rightarrow$ ① st.top()

② st.top right $\neq$ Null

$\downarrow$
leftmost nodes
push

10
50
100

SC: $O(n)$



-5, 0, 20, 25, 50, 75, 100, 150, 175, 200, 300

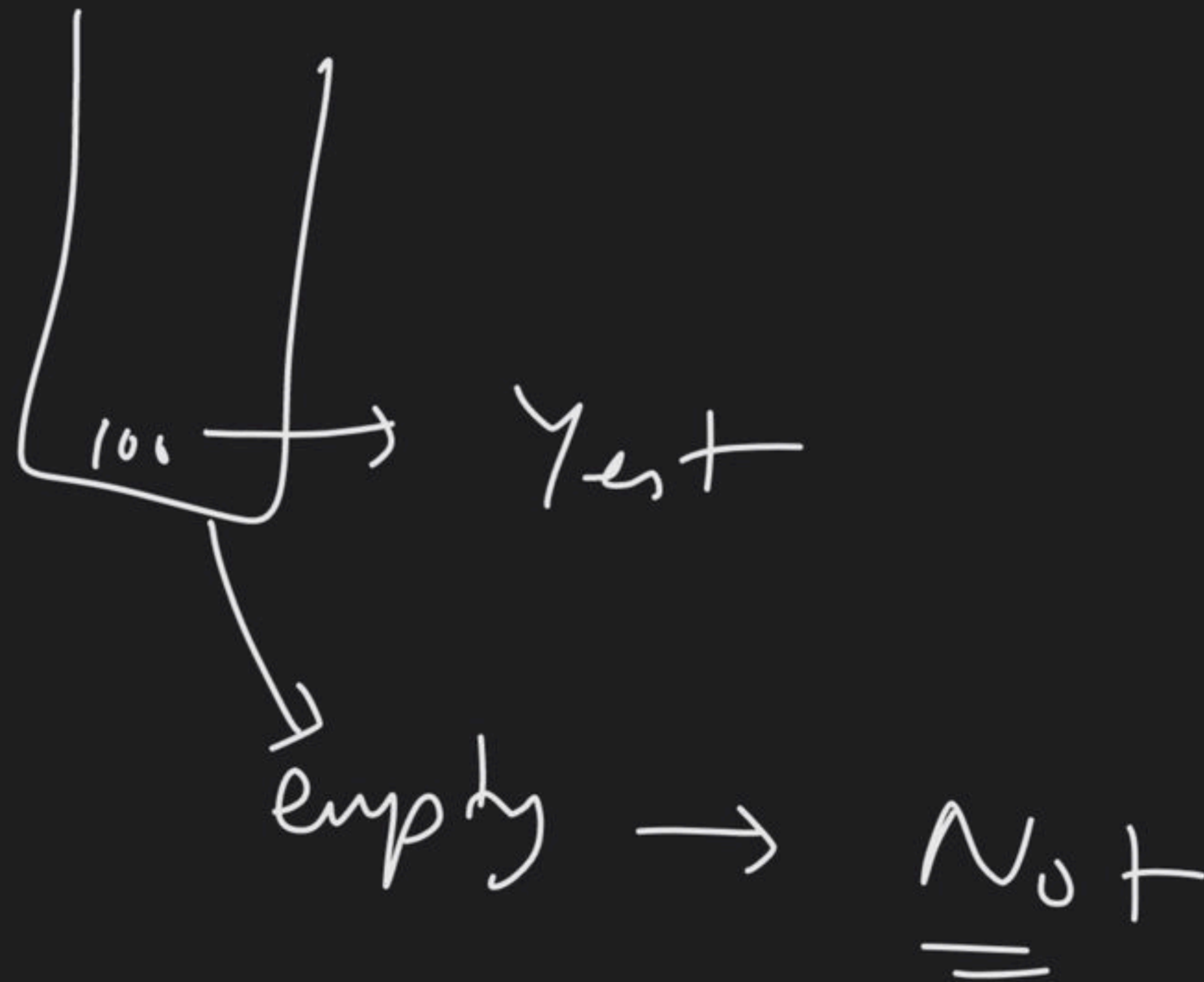
~~200~~

~~175~~

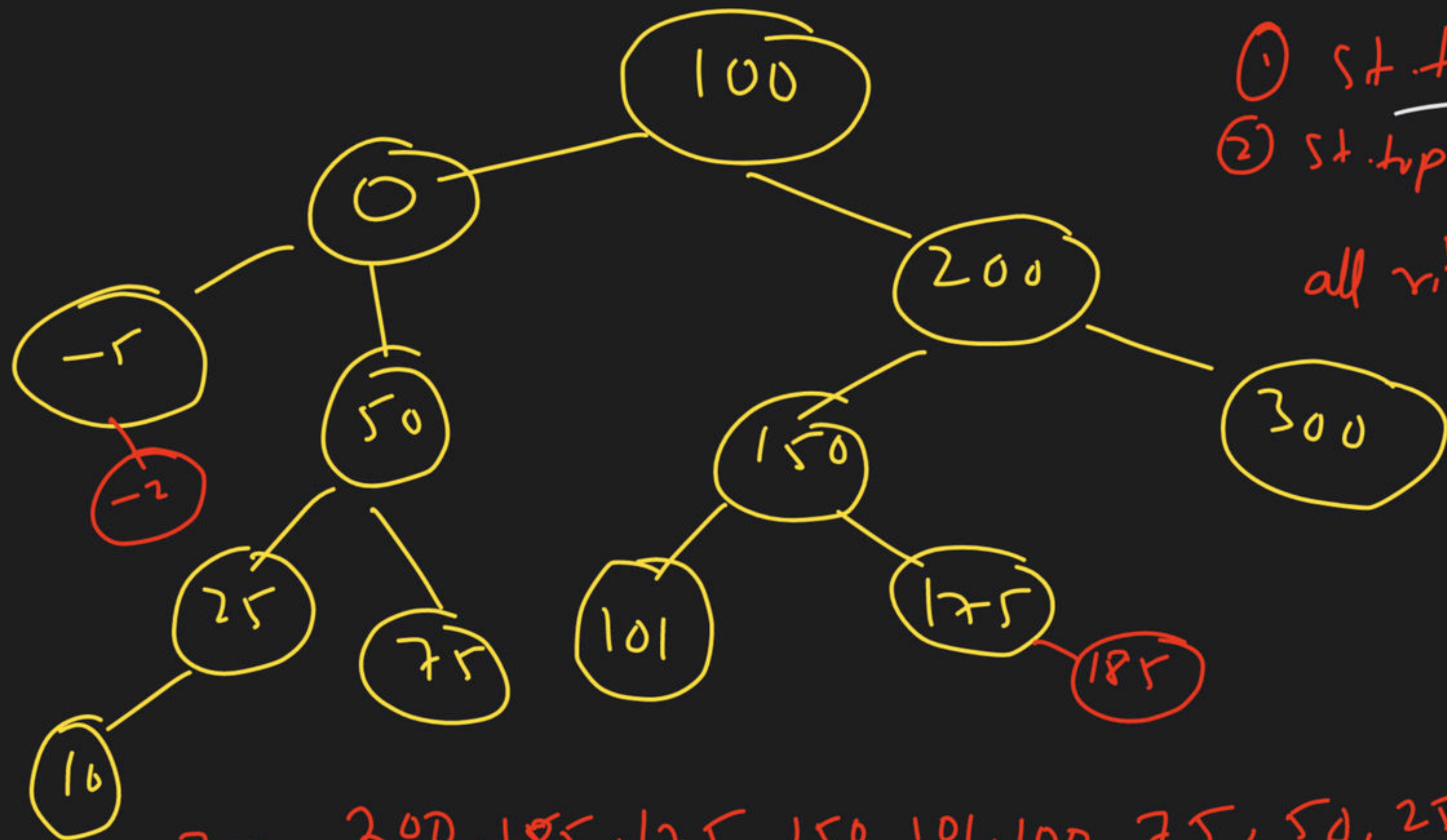
~~150~~

~~200~~

hasNext() ==>



Can I do Reverse inorder Traversal? Sorted Σ
 $\searrow$ desc

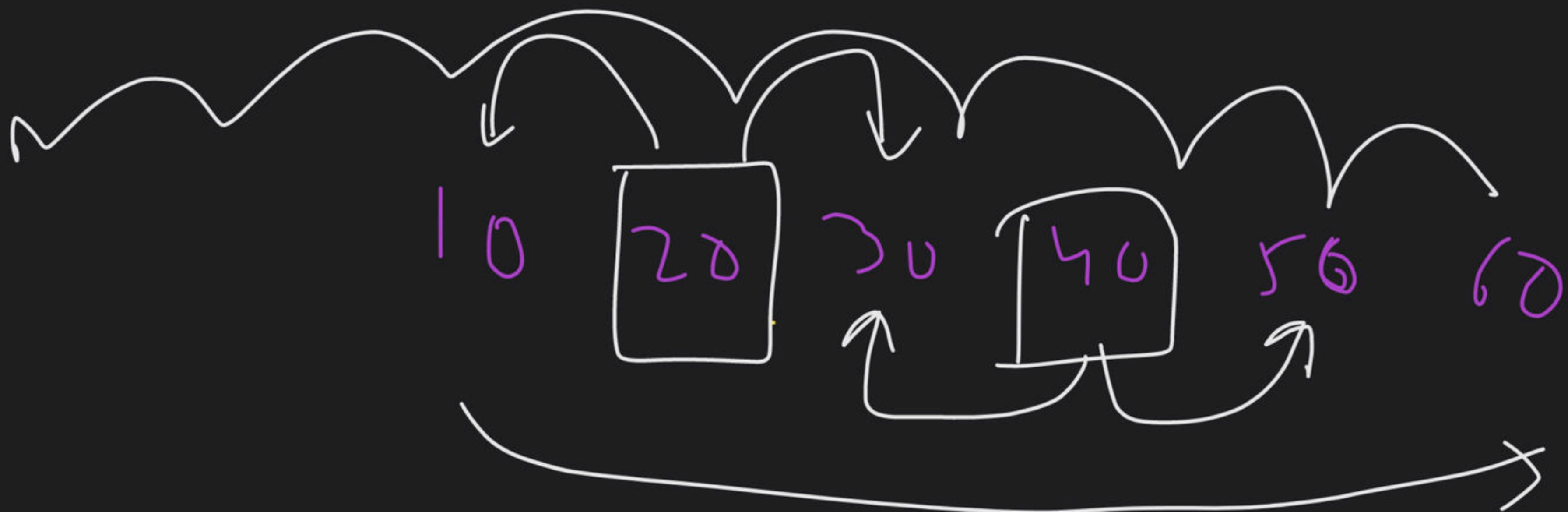


- ① st.top
- ② st.top left
 $\downarrow$
all right node push

300, 200, 185, 175, 150, 101, 100, 75, 50, 25, 10, 0
-2, -5

① before() $\Rightarrow$ left

② has Before() $\Rightarrow$

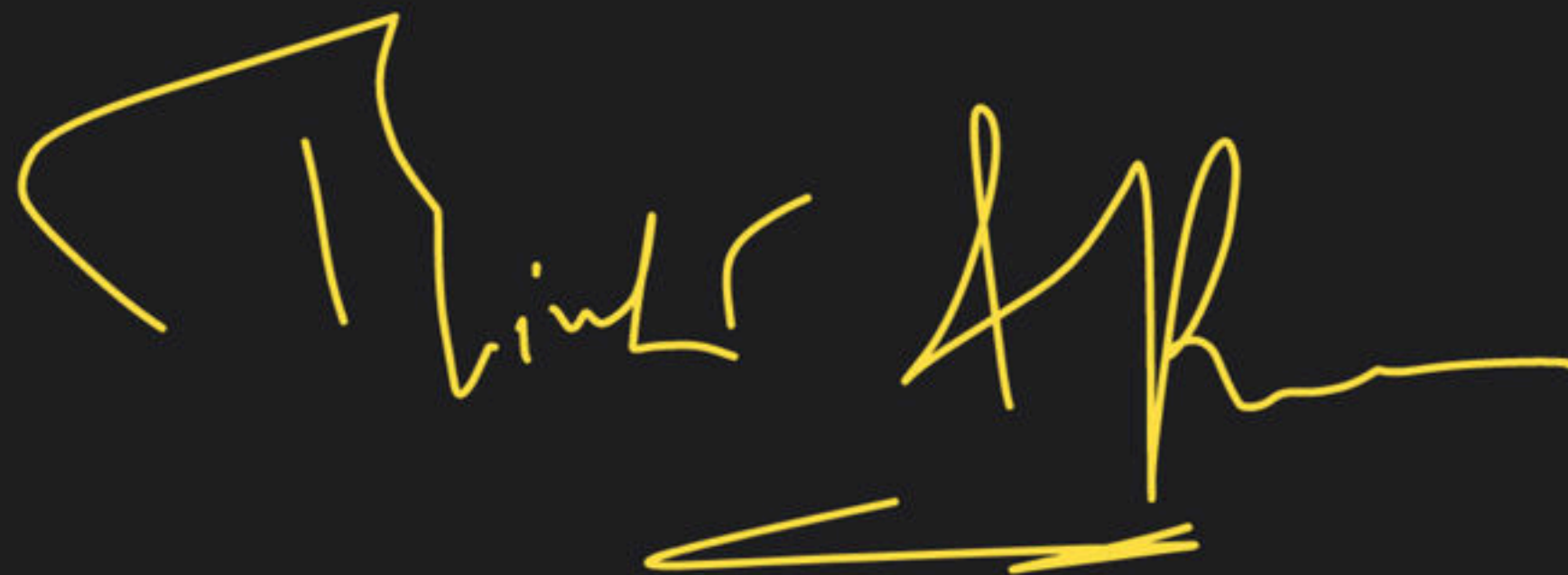


④

Two Sum IV $\rightarrow$ BST

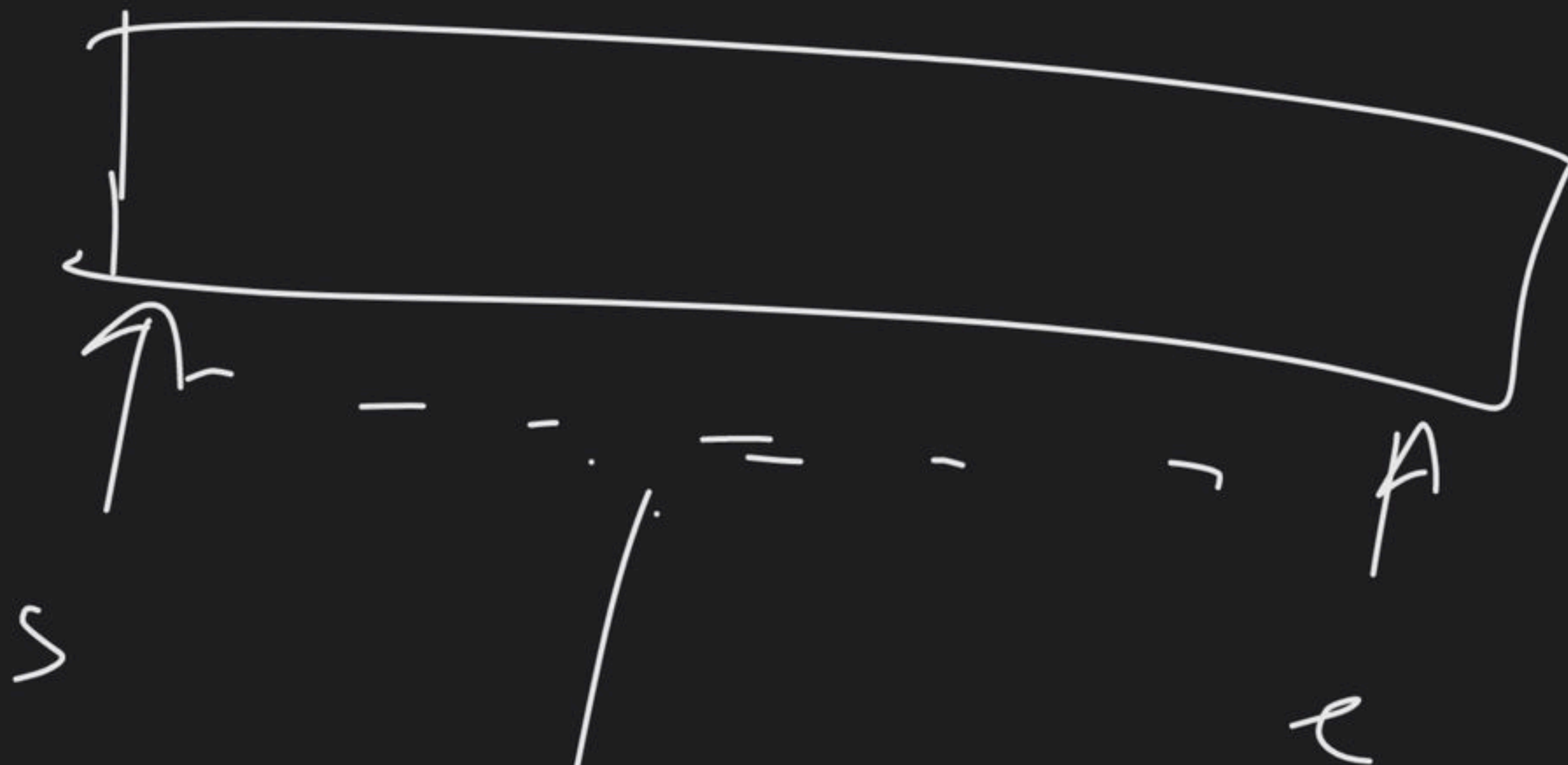
LC: 653 $\rightarrow$ 

9:47 PM =



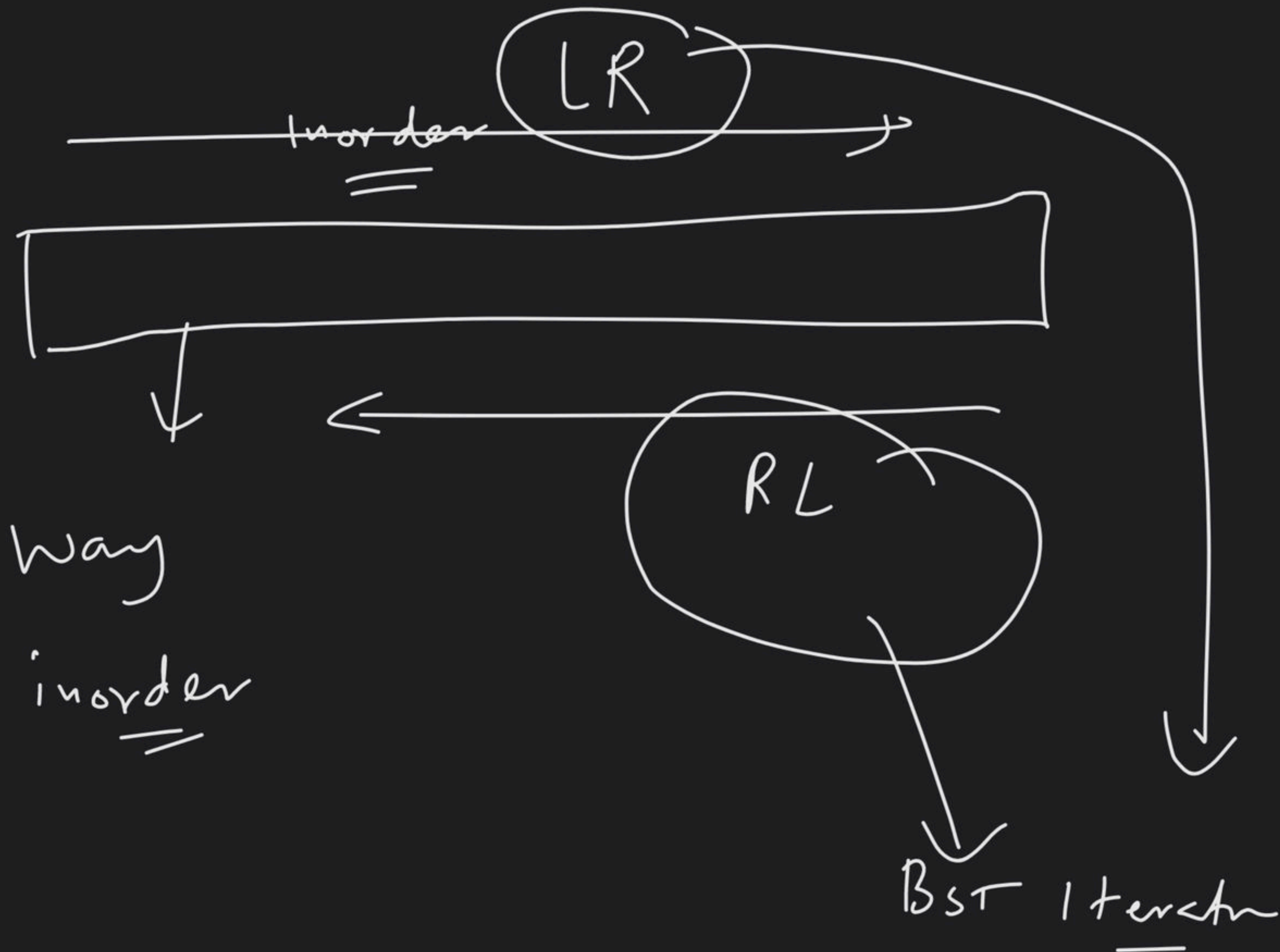
① Inorder →

Sorted.



Two Sum Problems

②

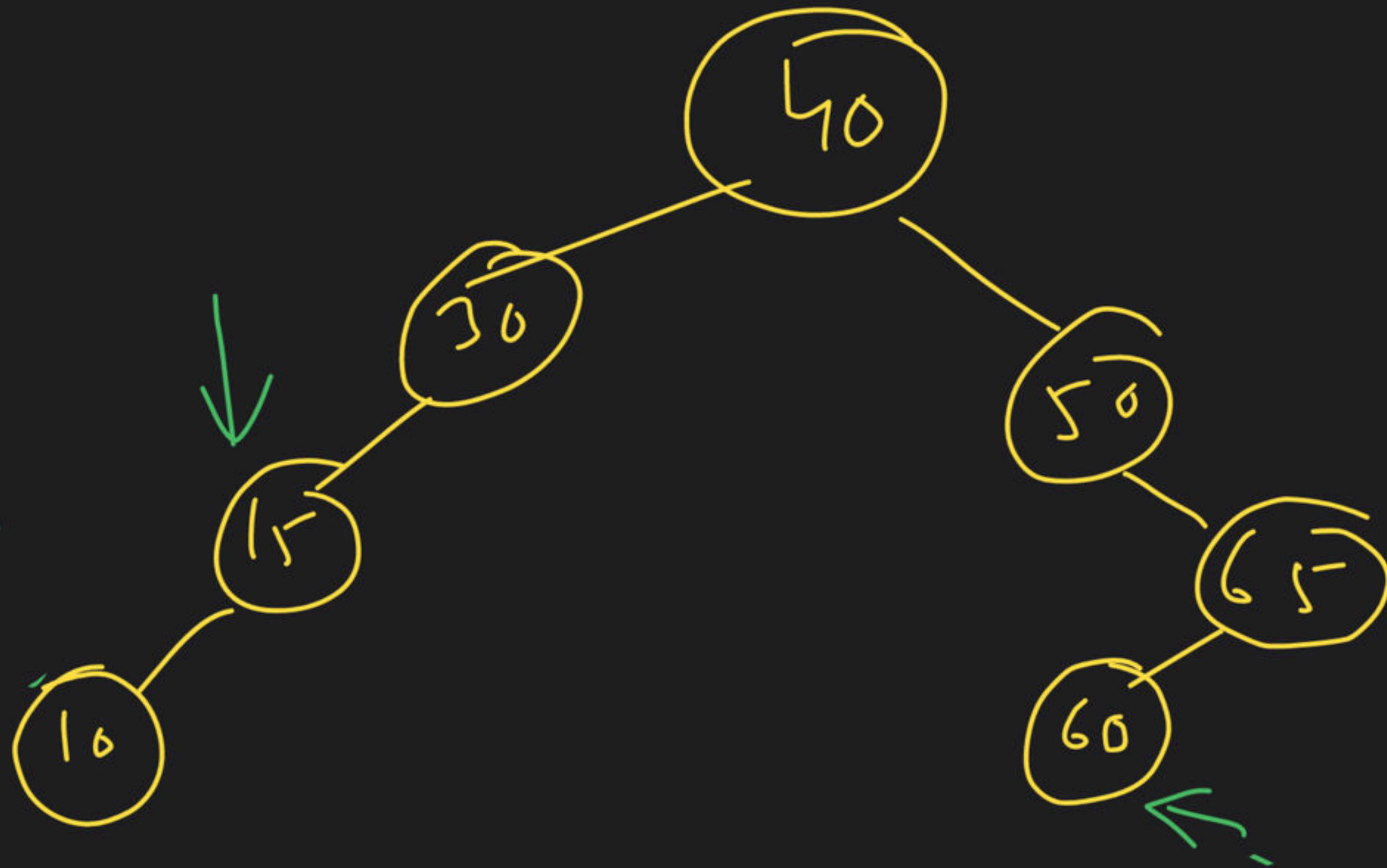


0 1 2 3 4 5 6
10 / 15 / 30 / 40 / 50 / 60 / 65
↑ s ↑ e

$$k = \textcircled{80}$$

next $\Rightarrow$ 10

Before = 65



⑤ LC: 99 Recover BST

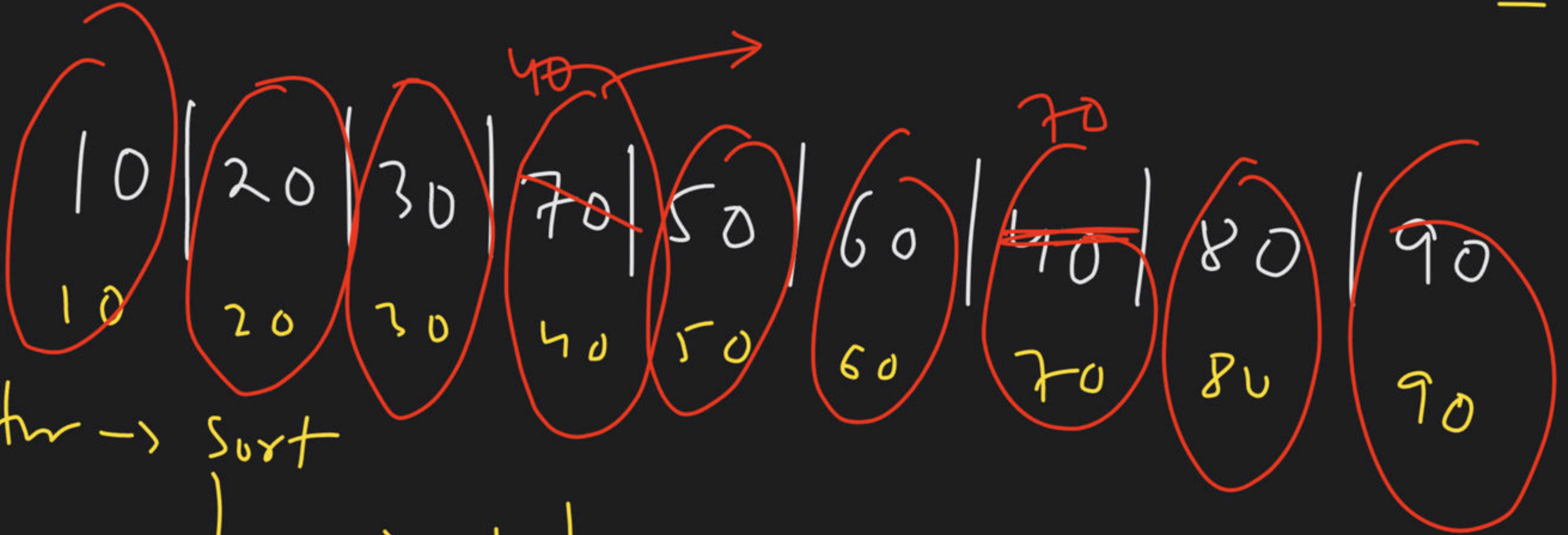
10:08 PM $\Rightarrow$ Think 4 Read

(M1)

Brute force

①

Inorder $\rightarrow$ vector



Vector $\rightarrow$ Sort

$\rightarrow$ 10 | 20

② Sort the vector  structure remain intact

③ again inorder traverse, & set

the values

$$TC \Rightarrow O(n \log n) \Rightarrow \underline{\underline{O(n) + O(n \log n) + O(n)}}$$
$$SC \Rightarrow O(n)$$

M2

Traverse karte karte solve ??

10 | 20 | 30 | 50 | 40 | 60 | 70 | 80 | 90 | 100

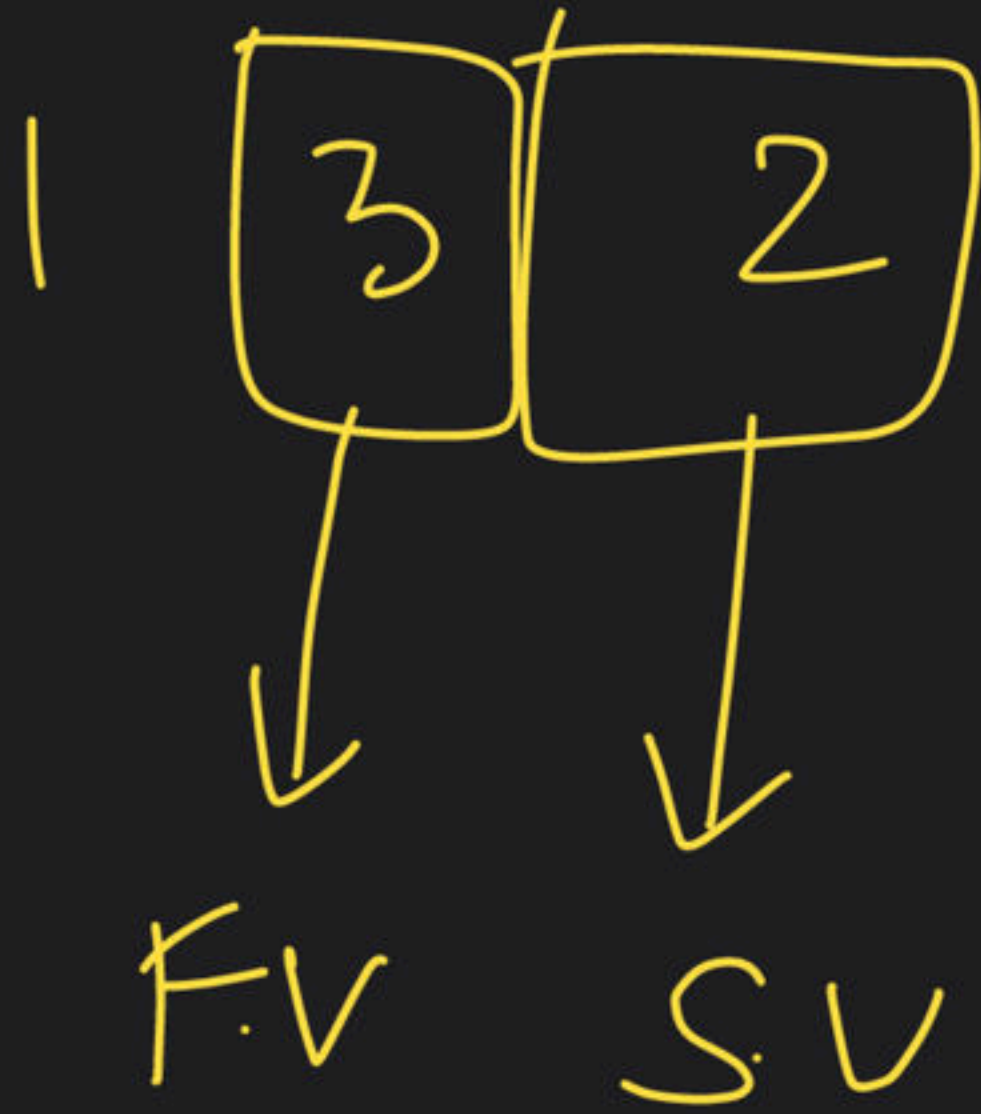
* > *

if (C → val < Prev → val)
// violation

if (F == 0) first = Prev;
Second = Current;

first = 50
Second = 40

→
adj



4

5

6

7

Case 2: Non adj. Violation node

1 2 6 4 5 3 7



F.V

S.V

will never update

Can update

10 | 20 | 30 | 80 | 50 | 60 | 70 | 40 | 90 | 100

AB

// store & compute

F.V = 80 ✓

S.V = ~~50~~

40 ✓

swap

if (curr->val < prev->val)
{
 if (F.V == Null)
 F.V = prev.

S.V = curr;

3